

Complete chloroplast DNA sequence of the moss *Physcomitrella patens*: evidence for the loss and relocation of *rpoA* from the chloroplast to the nucleus

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ABSTRACT

The complete chloroplast DNA sequence (122 890 bp) of the moss *Physcomitrella patens* has been determined. The genome contains 83 protein, 31 tRNA and four rRNA genes, and a pseudogene. Four protein genes (*rpoA*, *cysA*, *cysT* and *ccsA*) found in the liverwort *Marchantia polymorpha* and the hornwort *Anthoceros formosae* are absent from *P.patens*. The overall structure of *P.patens* chloroplast DNA (cpDNA) differs substantially from that of liverwort and hornwort. Compared with its close relatives, a 71 kb region from *petD* to *rpoB* of *P.patens* is inverted. To investigate whether this large inversion and the loss of *rpoA* usually occur in moss plants, we analyzed amplified cpDNA fragments from four moss species. Our data indicate that the large inversion occurs only in *P.patens*, whereas the loss of the *rpoA* gene occurs in all mosses. Moreover, we have isolated and characterized the nuclear *rpoA* gene encoding the α subunit of RNA polymerase (RNAP) from *P.patens* and examined its subcellular localization. When fused to green fluorescent protein, RpoA was observed in the chloroplasts of live moss protonemata cells. This indicates that chloroplast RNAP is encoded separately by chloroplast and nuclear genomes in the moss. These data provide new insights into the regulation and evolution of chloroplast transcription.

INTRODUCTION

Chloroplasts evolved from cyanobacteria through endosymbiosis, and possess independent genomes (1). To date, complete chloroplast DNA (cpDNA) sequences from two

dozen land plants and algae have been determined (2,3). Algal cpDNAs range from 89 (*Codium fragile*) to over 1500 kb (*Acetabularia*), and their gene content and organization vary from species to species (2). In contrast, land plant cpDNAs are relatively uniform in size, from 120 to 160 kb, and their gene content and organization are well conserved (3). Bryophytes consist of three classes (liverworts, hornworts and mosses) and have been identified as the earliest land plants (4–6). Among the bryophytes, the cpDNA (121 024 bp) of the liverwort *Marchantia polymorpha* was the first to be sequenced (7), and the hornwort *Anthoceros formosae* cpDNA sequence (161 162 bp) has recently been published (8). The larger size of the cpDNA genome of *A.formosae* results from a larger inverted repeat (IR) sequence compared with the liverwort *M.polymorpha*. The overall gene organization is highly conserved between the two bryophytes. Single introns, however, are inserted into the 23S rRNA genes (*rrn23*) of *A.formosae* (8) but not into those of *M.polymorpha* (7), the charophyte *Chaetosphaeridium globosum* (9) or vascular plants (3). Over 900 RNA editing sites have been identified in the hornwort *A.formosae* chloroplast (10). In contrast, no RNA editing site has been found in the cpDNA of the liverwort *M.polymorpha* (11). Therefore, RNA editing events differ substantially between bryophyte species compared with the high degree of conservation of editing sites in the chloroplasts of vascular plants (12). Significant divergence in RNA editing events among the bryophytes supports bryophyte paraphyly, as suggested by molecular phylogenetic studies (13–15).

The remaining bryophyte group, the mosses, has been reported to be most closely related to the early lycopod land plants, the ferns, and the vascular plant lineage (4–6,15). The exact phylogenetic status of this group, however, remains unresolved. To answer this question, molecular data on moss chloroplast genes are required. There is little information on the gene structure (16–18) or RNA editing (19) of the moss *P.patens*. Therefore, we have determined the complete nucleotide sequence of the *P.patens* cpDNA. Moreover, we

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have isolated a nuclear *rpoA* gene encoding an α subunit of RNA polymerase (RNAP) and discuss the relocation of *rpoA* from the chloroplast to the nucleus.

MATERIALS AND METHODS

Accession numbers

The sequences reported in this paper have been deposited in the DDBJ/GenBank/EMBL databases with accession numbers: *P.patens* cpDNA, AP005672; cpDNA fragments amplified from four mosses, AB098724–AB098727; PpRpoA gene, AB110071; PpRpoA cDNA, AB110072.

Plant materials

The moss *P.patens* subspecies *patens* was grown at 25°C under continuous illumination. Total cellular DNA and RNA were isolated from *P.patens* protonema, as described previously (20).

Preparation of cpDNA fragments

A *P.patens* genomic library was screened using the tobacco *ndhD* sequence (21) as a probe, as described previously (22), and the genomic clone λ -D5 was isolated. The λ -D5 insert DNA was amplified by polymerase chain reaction (PCR) using primers designed from λ fix-II vector sequences, AGC-TCTAATACGACTCACTATAGGGCGTCTGA and GAGCT-CAATTAACCCTCACTAAAGGGAGTCTGA. The reaction of long and accurate (LA) PCR was performed with 30 cycles of 10 s denaturation at 98°C and 15 min annealing and extension at 68°C using LA *Taq* DNA polymerase (Takara Shuzo). The amplified DNA contained a 12 640 bp region from *rpl32* to *trnV-GAC*. To further amplify cpDNA fragments, total cellular DNA (90 ng) was subjected to LA PCR under the same conditions as above using primers designed from either nucleotide sequences deposited in DNA databases or sequences determined in this study as follows: for amplification of a 19 379 bp DNA region from *psbD* to *rbcL*, LA1 (TTAGGAGGTCTATGGACTTTCGTTGCT-CTT, AB013655) and LA2 (AGTCATCACGAAGTAAGT-CAACAAACCCTA, AB066297); for a 14 473 bp DNA region from *psbB* to *rbcL*, LA3 (TAAGTAAAAAAGGATGATGGAAAA, this study) and LA4 (AGGTTCTGT-TACTAACTTATTTACTTCTATTG, AB066297); for a 13 774 bp DNA region from *psbB* to *rrn16*, LA5 (AGTAT-TGCTGCTGTATTGTTTGCTTTT, this study) and LA6 (TTTGAGTTTCATTCTTGCGAACGTACTCCC, this study); for a 17 834 bp DNA region from *rpl21* to *rrn4.5*, LA7 (TGACGCTATAATTGAAACCGGAGGTGAAC, this study) and LA8 (TTTATCFATCACGATAGGTGCCAAGTGGAAG, this study); for a 15 083 bp DNA region from *rpoC1* to *rrn16*, LA9 (GATGATTTTTAATTGTTAGTATGTA-TAGTCC, AB013657) and LA10 (TTTGAGTTTCATTC-TTGCGAACGTACTCCC, this study); for a 9799 bp DNA region from *psbA* to *psbD*, LA11 (TGTAGGTATTTGGTT-TACTGCTTTAGGTATC, X04465) and LA12 (ACCAAC-TACTCCAATAGCACTCATCCATAAA, AB013655); for a 19 156 bp DNA region from *trnH* to *rpoC1*, LA13 (AAATAATAAAAAATGGGCGAACGACGGGAAT, this study) and LA14 (AAAATCATCAAGGTATCTATGGTAA-TAAAAA, AB013657); for an 11 365 bp DNA region from

rps7 to *chlL*, LA15 (GTTTCTTCTTTTTTCGTATTGCT-TCTCCAC, this study) and LA16 (GTAAATAACAT-CTTCAGGCCAAACATCTTCAT, this study). The eight amplified cpDNA fragments and the λ -D5 insert DNA encompassed the entire *P.patens* chloroplast genome.

Sequence analysis

The amplified cpDNA fragments were sheared, cloned into pUC18, and shotgun-sequenced using Shimadzu RISA384 sequencers by the Department of Genomic Research, Shimadzu Co., Japan. Nucleotide sequence files were assembled using the Phrap program (Phil Green, University of Washington, Seattle, WA, USA) and the resulting sequences were analyzed with Genetyx-Mac v.9.0 (Software Development) and Sequencer software (Hitachi Software Engineering). When the nucleotide sequences of either protein- or RNA-coding regions contained ambiguous bases or apparent small gaps, they were corrected by sequencing of re-amplified *P.patens* DNA fragments (19,23). Fifteen cpDNA regions with small gaps and 33 protein- or RNA-coding regions were amplified from total cellular DNA, and subcloned into pGEM-T Easy (Promega). At least three independent clones were picked up and sequenced using the DYEnamic ET Terminator Cycle Sequencing Kit (Amersham Bioscience) and an ABI PRISM 3100 DNA sequencer (Perkin-Elmer, Applied Biosystems). Database searches and sequence data analyses were performed as described previously (19).

DNA analysis of four mosses

Total cellular DNA from the mosses *Hylocomium splendens* (Hedw.) Bruch and Schimp., *Plagiothecium eurphyllum* (Card. and Ther.) Iwats., *Bartramia pomiformis* Hedw. and *Ceratodon purpureus* (Hedw.) were used as templates for PCR. Primers P1–P4 were designed from the sequences that have been determined for the *P.patens* genes *rps11*, *rpoB*, *petD* and *petN*, respectively: P1, 5'-TTTTGTTCGTGATG-TAACTCCTATG-3'; P2, 5'-CTACCATAGCATCCTCAG-TAGATT-3'; P3, 5'-CTAAATTAGCTAAAGGTATGG-GTC-3'; and P4, 5'-TAAATCTAATTTTTATAATCCGC-TTC-3'. The amplified DNA fragments were subcloned into pGEM-T Easy (Promega) and sequenced as described previously (19,24).

Isolation and sequence analysis of cDNA and genomic DNA clones

Full-length cDNA encoding PpRpoA was prepared by 5'-rapid amplification of cDNA ends using primers designed from an expressed sequence tag (EST) clone as described (24). Genomic clones were isolated using the cDNA probe from a *P.patens* genomic library (19). Sequencing was performed with an ABI PRISM 3100 sequencer and the DYEnamic ET Terminator Cycle Sequencing Kit (Amersham Bioscience) using appropriate sequencing primers.

Construction of fusion genes and microscopic observation

A DNA fragment encoding the N-terminal 94 amino acid residues of PpRpoA was amplified from the cDNA as above with oligonucleotides containing restriction sites (underlined), 5'-ATAGCCATGGCCTCGTCGACAGGAGCATC-3' and

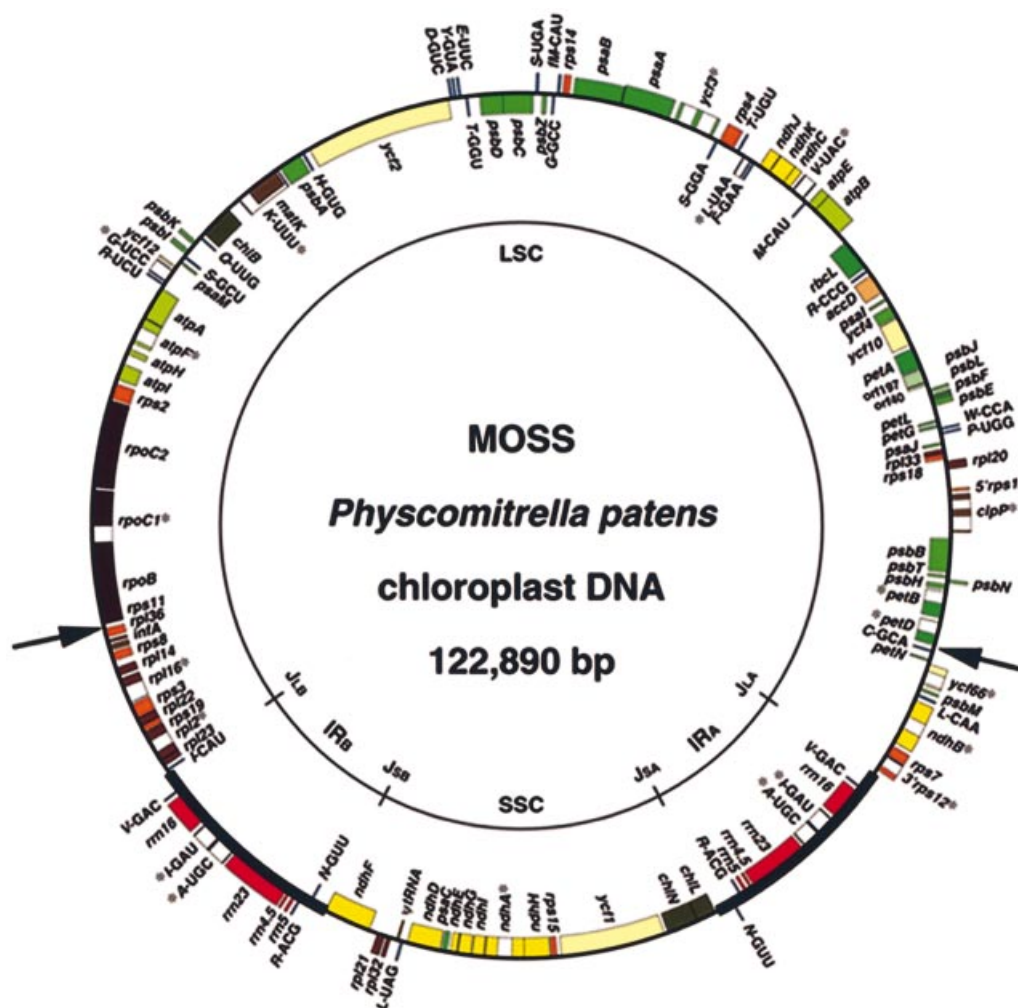


Figure 1. Gene map of the moss *P.patens* chloroplast genome. IR sequences (IR_A and IR_B) are separated by the SSC and LSC regions. Genes on the inside of the map are transcribed clockwise and genes on the outside are transcribed counter-clockwise. Genes with related functions are shown in the same color. Asterisks denote intron-containing genes or split genes. Arrows indicate positions of the endpoint of large inversion relative to liverwort *M.polymorpha* and hornwort *A.formosae* cpDNA.

5'-AGCTTCTAGAATGGCAGCAGTCATGAGCGC-3'. SalI-NcoI-digested amplified DNA fragments were introduced into SalI-NcoI-cleaved CaMV35S-sGFP(S65T)-nos3' (23) to create the *PpRpoA-gfp* construct. The reporter construct was introduced into the protoplasts prepared from *P.patens* protonemata by polyethyleneglycol-mediated transformation (20). One day after transformation, the fluorescence of green fluorescent protein (GFP) was monitored using a BX-50 fluorescence microscope (Olympus).

RESULTS

Overall structure of *Physcomitrella* cpDNA

The circular cpDNA of *P.patens* is 122 890 bp long, and its gene map is shown in Figure 1. Overall GC content of the cpDNA is 28.5%, which is closer to those of the liverwort *M.polymorpha* (28.8%) (7) and the hornwort *A.formosae* (32.9%) (8), the charophyte *Chaetosphaeridium* (29.6%) (9) and the algae (30.4–33%) (2), than to the GC content of the

vascular plants (38–39%) (3). Two identical copies of a large sequence (9589 bp) containing the rDNA operon (*rrn16–rrn23–rrn4.5–rrn5*) are present in an inverted orientation (IR_A and IR_B). A large single-copy (LSC) region of 85 212 bp and a small single-copy (SSC) region of 18 501 bp separate these sequences from one another. This quadripartite structure of cpDNA is well conserved in the bryophytes (7,8), a charophyte (9) and most vascular plants (3). BglII, ClaI and SacI restriction maps predicted from the complete nucleotide sequence coincide with the physical maps constructed previously (25).

Relative to hepatic bryophytes (*M.polymorpha* and *A.formosae*), an extensive rearrangement of the *P.patens* cpDNA is evident within the LSC region. A large inversion is present that encompasses a 71 kb region from *petD* to *rpoB*, its apparent endpoints lying between *rps11* and *rpoB* at one end and *petD* and *trnC-GCA* at the other (Fig. 1, arrows). To investigate whether this large inversion occurs in other moss species, we used PCR with different combinations of primers on the DNAs of *C.purpurea*, *P.eurphyllum*, *H.splendens* and

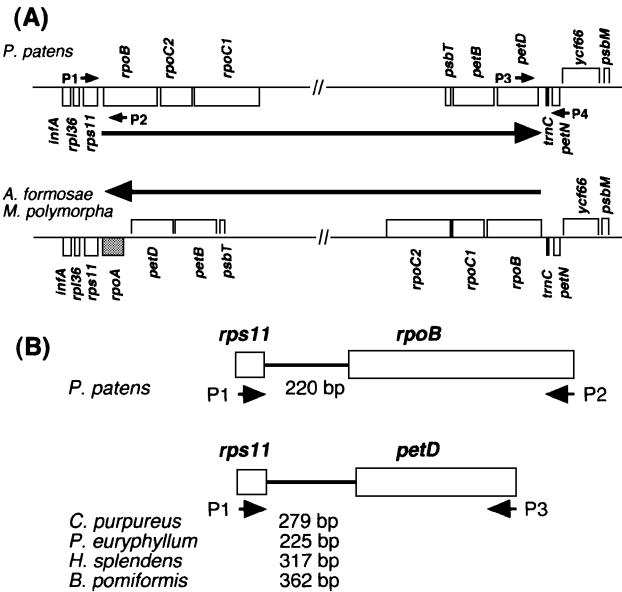


Figure 2. (A) Representation of the strategy used to detect the orientation of the 71 kb inversion. The *P. patens* (top), *M. polymorpha* and *A. formosae* (bottom) genomes are shown in the region of the 71 kb inversion. The positions of primers P1–P4, located adjacent to the inversion endpoints, are indicated. (B) DNA regions amplified with either the P1 and P2 pair or the P1 and P3 pair are illustrated, with the length (bp) of the intergenic spacer.

B. pomiformis. The combinations of primers P1 and P2, and primers P3 and P4 amplified the expected DNA fragments of 1002 and 1235 bp, respectively, from *P. patens* DNA (data not shown). In contrast, these primer combinations failed to amplify any DNA fragments from the DNA of other mosses. However, DNA fragments containing *rps11* and *petD* (Fig. 2B), or *rpoB* and *trnC* (data not shown), were amplified when different primer combinations (P1 and P3 or P2 and P4)

were used. This indicates that the large inversion occurs only in *P. patens*.

Gene content

One hundred and eighteen identified genes are listed in Table 1. Eighty-three protein-coding genes, including the hypothetical chloroplast reading frame (ycfs), were identified. The differences in genes in the cpDNAs of the three classes of bryophyte are summarized in Table 2. Four protein genes, *rpoA*, *cysA*, *cysT* and *ccsA* (*ycf5*) present in the liverwort *M. polymorpha* (7) and the hornwort *A. formosae* cpDNA (8), are absent from *P. patens* cpDNA. Two genes, *cysA* and *cysT*, that encode sulfate transport proteins, are present in the green algae *Chlorella* (26) and *Nephroselmis* (27), but are absent in charophytes (9) and vascular plants (4). *ccsA* is usually found in land plants (28–30) and algae (2,26), except *Euglena gracilis* (31). Furthermore, two open reading frames (ORFs) encoding 197 and 40 amino acid residues are located downstream of *petA*. The 70 N-terminal amino acid residues predicted from ORF 197 show ~30% identity with those of *Chaetosphaeridium* ORF 231 (9) and *Mesostigma* ORF 167 (32), which also occur downstream of *petA*. However, no sequences similar to ORF 197 are found in the cpDNAs of other plants or algae. A sequence homologous to ORF 40 (~40% amino acid identity) has been found in the hornwort *A. formosae* (8).

Thirty-five RNA genes were identified, of which 31 are tRNA genes and four are rRNA genes. A tRNA-like sequence is present between *rpl32* and *trnL-UAG* in the SSC region (Fig. 1). The nucleotide sequence shows 71.6% identity with that of *A. formosae* *trnP*-GGG (8) and 62.1% with that of *M. polymorpha* pseudo *trnP*-GGG (7), which are located in similar positions to the *P. patens* tRNA-like sequence. However, the *P. patens* sequence encodes a tRNA with an AAC anticodon, which is complementary to the GTT codon (encodes valine). No tRNAs or tRNA genes with AAC

Table 1. Genes encoded by *P. patens* cpDNA

Gene products	Genes
Photosystem I	<i>psaA</i> , B, C, I, J, M
Photosystem II	<i>psbA</i> , B, C, D, E, F, H, I, J, K, L, M, N, T, Z
Cytochrome <i>b6/f</i>	<i>petA</i> , B ^a , D ^a , G, L, N
ATP synthase	<i>atpA</i> , B, E, F ^a , H, I
Chlorophyll biosynthesis	<i>chlB</i> , L, N
Rubisco	<i>rbcL</i>
NADH oxidoreductase	<i>ndhA</i> ^a , B ^a , C, D, E, F, G, H, I, J, K
Large subunit ribosomal proteins	<i>rpl2</i> ^a , 14, 16 ^a , 20, 21, 22, 23, 32, 33, 36
Small subunit ribosomal proteins	<i>rps2</i> , 3, 4, 7, 8, 11, 12 ^{a,b} , 14, 15, 18, 19
RNAP	<i>rpoB</i> , C1 ^a , C2
Translation factor	<i>infA</i>
Other proteins	<i>accD</i> , <i>clpP</i> ^c , <i>matK</i>
Proteins of unknown function	<i>ycf1</i> , 2, 3 ^c , 4, 10, 12, 66 ^a
Ribosomal RNAs	<i>rrn16</i> ^d , 23 ^d , 4.5 ^d , 5 ^d
Transfer RNAs	<i>trnA</i> (UGC) ^{a,d} , C(GCA), D(GUC), E(UUC), F(GAA), G(GCC), G(UCC) ^a , H(GUG), I(CAU), I(GAU) ^{a,d} , K(UUU) ^a , L(CAA), L(UAA) ^a , L(UAG), fM(CAU), M(CAU), N(GUU) ^d , P(UGG), Q(UUG), R(ACG) ^d , R(CCG), R(UCU), S(GCU), S(GGA), S(UGA), T(GGU), T(UGU), V(GAC) ^d , V(UAC) ^a , W(CCA), Y(GUA)

^aGene containing a single intron.

^bGene divided into two independent transcription units.

^cGene containing two introns.

^dTwo gene copies due to the IR.

Table 2. Gene content of cpDNAs from green alga, charophyte, bryophytes and land plants

Plants	<i>rpoA</i>	<i>ccsA</i>	<i>cysA</i>	<i>cysT</i>	<i>ycf66</i>	<i>matK</i>	<i>rps15</i>	<i>trnP-GGG</i>
<i>Chlorella</i> (26)	+	+	+	+	–	–	–	–
<i>Chaetosphaeridium</i> (9)	+	+	–	–	+	+	+	+
<i>Marchantia</i> (7)	+	+	+	+	+	+	+	ψ
<i>Anthoceros</i> (8)	+	+	+	+	–	ψ	ψ	+
<i>Physcomitrella</i>	–	–	–	–	+	+	+	–
<i>Psilotum</i> (28)	+	+	–	–	–	+	+	+
<i>Pinus</i> (29)	+	+	–	–	–	+	+	+
<i>Arabidopsis</i> (30)	+	+	–	–	–	+	+	–

The presence (+) or absence (–) of each molecular character, and pseudogene (ψ) are shown. References are shown in parenthesis.

anticodons have been found in other organelles or cyanobacteria. Therefore, we tentatively classified this tRNA-like sequence as a pseudogene. Genes encoding stable RNAs other than rRNAs and tRNAs have been identified for *tscA* in *Chlamydomonas reinhardtii* (33), for *rnpB* and *tmRNA* genes in the non-green algae (34), and for *sprA* in tobacco (35), but sequences homologous to these genes are not present in this moss.

Intron content

Eighteen genes for six tRNAs and 12 proteins contain introns, as shown in Figure 1 and Table 1. The intron of the *trnL* gene encoding tRNA^{Leu}-UAA is classified into the group I-type intron and the remaining introns of the 17 genes are categorized into the group II intron (8,21). A maturase-like polypeptide is encoded within the longest intron (2168 bp) of *trnK* coding for tRNA^{Lys}-UUU. The gene for ribosomal protein S12 is divided into 5'-*rps12* and 3'-*rps12*, and each gene segment can be transcribed independently and the transcripts *trans*-spliced. *clpP* and *ycf3* contain two introns, as in the corresponding tobacco and rice genes, and the other intron-containing genes have a single intron. *Anthoceros formosae ycf3* also has two introns, whereas *M.polymorpha ycf3* has only the first of these introns. *ycf66* is absent in *A.formosae*, but present in *P.patens* and *M.polymorpha*. The intron of *rrn23* is present in *A.formosae* as well as in the green algae *Chlorella* (26) and *Chlamydomonas* (36), but not in *P.patens*, *M.polymorpha* (8) or the prasinophytes *Mesostigma* (32) and *Nephroselmis* (27), which constitute the most ancient green-plant lineage (5).

Loss of *rpoA* from moss cpDNA

The *rpoA* gene is normally present immediately downstream from a ribosomal-protein gene cluster (*rpl23* to *rps11*) and is located between *rps11* and *petD* in most land plants. In contrast, *P.patens rpoB* is located downstream from *rps11*, and *rpoA* is completely absent from the cpDNA (Fig. 1). Interestingly, in the other moss species, *H.splendens*, *P.euryphyllum*, *B.pomiformis* and *C.purpureus*, *petD* is located downstream from *rps11*, and there is no *rpoA* gene located between them (Fig. 2B). This observation strongly suggests that the loss of the *rpoA* gene is a general occurrence in mosses. However, the possibility that the *rpoA* gene is present at some other locus on the cpDNA cannot be ruled out for other mosses.

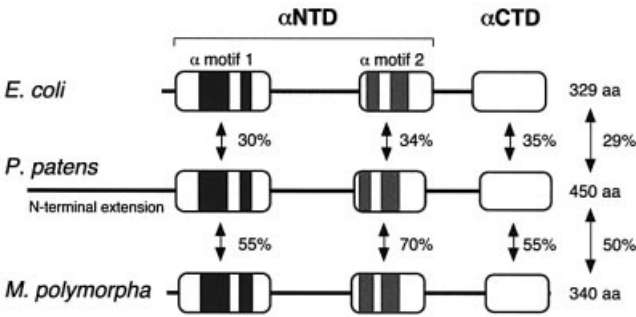


Figure 3. Schematic diagram showing the domain structure of the α subunit of *E.coli* RNAP, *P.patens* nuclear PpRpoA and *M.polymorpha* chloroplast RpoA. α NTD and α CTD denote α subunit N-terminal and C-terminal domains, respectively. The gray boxes denote regions conserved in sequence between α homologs of prokaryotic, archaeobacterial, chloroplast and eukaryotic RNAPs.

Identification of the nuclear *rpoA* gene

The absence of *rpoA* in the moss cpDNA strongly suggests the transfer of the *rpoA* gene from the chloroplast to the nuclear genome. To isolate the nuclear *rpoA* counterpart, we searched the DNA databases and identified an EST (GenBank accession no. BI740521) encoding the C-terminal 57 amino acid residues of the α subunit. We next obtained and sequenced the full-length cDNA. The predicted protein contains 450 amino acid residues with the characteristic arrangement of protein domains identified in the *Escherichia coli* α subunit of RNAP (37) and shows 50% amino acid identity with chloroplast-encoded liverwort *M.polymorpha* RpoA and 29% identity with the *E.coli* α subunit (Fig. 3). Therefore, we designated this protein and gene PpRpoA and *PpRpoA*, respectively. The *P.patens* α homolog contains an N-terminal extension, which could be a transit peptide for targeting this protein to chloroplasts. The *PpRpoA* gene consists of two exons, which are separated by a 215 bp intron. The first exon encodes only the 94 amino acid N-terminal extension that is predicted to target PpRpoA to the chloroplast. Genomic Southern analysis showed that PpRpoA is encoded by a single copy gene (data not shown).

Chloroplast localization of the moss PpRpoA

The N-terminal sequence of PpRpoA was predicted to specify chloroplast targeting (a score of 0.911) by the TargetP

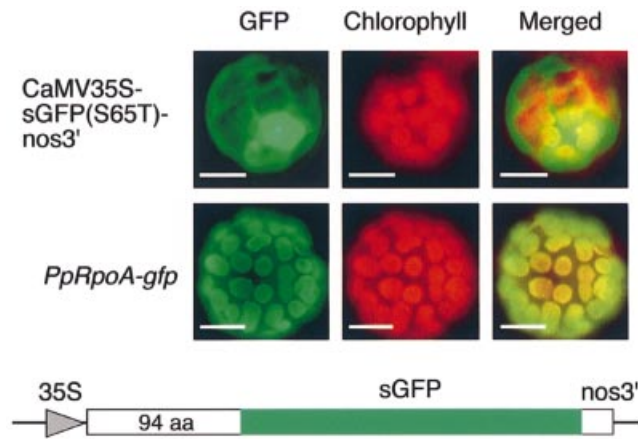


Figure 4. Localization of fusion protein encoded by the *PpRpoA-gfp* construct to chloroplasts. The *PpRpoA-gfp* was introduced into the *P.patens* protoplasts of protonemata. The localization of GFP and chloroplast pigments (chlorophyll) in transformed cells was detected by fluorescent microscopy. Merged images are shown on the right. Bars are 10 μ m.

program for protein sorting (38). To localize PpRpoA in cells we further investigated the cellular localization in protoplasts of *P.patens* protonemata through the use of a chimeric protein comprised of the PpRpoA N-terminal 94 amino acid residues fused to GFP. The *PpRpoA-gfp* construct produced GFP fluorescence associated with chloroplasts (Fig. 4). This clearly demonstrated that the N-terminal extension of the PpRpoA functions as a chloroplast-targeting signal. This strongly suggests that the nuclear-encoded PpRpoA is a component of chloroplast RNAP.

DISCUSSION

The size and quadripartite structure of the chloroplast genome of *P.patens* is very similar to that of *M.polymorpha*. The cpDNA of *P.patens*, like the vascular plants, however, lacks *cysA* and *cysT* and contains a *ycf3* gene with two introns, whereas that of the liverwort does not. This implies that the moss *P.patens* diverged from the hepatic bryophytes and is more closely related to the vascular plants.

The most striking feature of the cpDNA of the moss *P.patens* is the absence of *rpoA*. After the divergence of the mosses from the hepatic bryophytes, *rpoA* was lost from the cpDNA, together with *ycf5*, *cysA* and *cysT*. The loss of *rpoA* gene appears to be typical in mosses, and poses the question as to whether moss chloroplasts possess a transcription system unique among the green land plants. *rpoA* is already known to be absent from cpDNAs of the photosynthetic plant geranium (*Pelargonium hortorum*) (39), the parasitic vascular plant *Epifagus virginiana* (40), and from the plastid-like genome of the malaria parasite *Plasmodium falciparum* (41). The retained 60% of *rpoA* in the geranium cpDNA is extensively fragmented but an intact *rpoA* gene was not detected in the nuclear DNA by Southern hybridization analysis (39). In contrast, *rpoB*, *C1* and *C2* are most likely intact genes in *P.hortorum* (39) and *P.falciparum* (41). This suggests that *rpoA* gene was preferentially lost from the cpDNA, rather than *rpoB*, *C1* and *C2*. The loss of single genes or groups of functionally related genes occurred similarly in specific

clades, e.g. the loss of *ycf1*, *ycf2* and *accD* in monocotyledonous plants (42,43), the loss of all *ndh* genes in the gymnosperm *Pinus* (29) and the loss of *infA* during angiosperm evolution (44).

In this study we have identified a nuclear *rpoA* gene encoding a protein predicted to contain 450 amino acids with high sequence identity to the α subunit of *E.coli* RNAP and the cpDNA-encoded RpoA (Fig. 3). Our findings clearly indicate that a functional *rpoA* has been transferred from the chloroplast to the nucleus. This is the first evidence for the presence of a nuclear *rpoA*. Nuclear *rpoA* genes with a chloroplast transit sequence are not found in *Arabidopsis thaliana* (45) and rice (46), whose cpDNAs retain an intact *rpoA*.

There are 171 identical amino acid residues between PpRpoA and liverwort *M.polymorpha* RpoA, of which 98 are encoded by codons that differ at synonymous positions. For example, *PpRpoA* preferentially uses GC residues at the third position in Leu, and generally uses the codons TTC for Phe, AAG for Lys and GAG for Glu. These codons are also preferentially used in *P.patens* ESTs (18). In contrast, A or T residues are preferentially used for the *M.polymorpha rpoA* as well as *P.patens* chloroplast protein genes. This indicates that the transfer of the moss *rpoA* to the nucleus was not a recent event and may have occurred immediately after the divergence of mosses from the hepatic bryophytes. There is also an EST (AW098196) in the databases from the moss *C.purpureus* which is predicted to encode a polypeptide similar to the *P.patens* PpRpoA. This indicates that mosses probably possess in general a transferred nuclear *rpoA* gene.

The α subunit is well characterized to be essential for RNAP assembly and basal transcription in *E.coli* (47,48). Therefore, in the moss chloroplast, the α subunit is indispensable for the function of plastid-encoded plastid RNAP (PEP), and the PpRpoA probably functions as the α subunit of PEP. Many *P.patens* chloroplast genes have canonical -10 and -35 sequences similar to those of *E.coli* promoter sequences (49). Moreover, cDNAs encoding σ -like transcription initiation factor for PEP have been isolated from *P.patens*, and the encoded transit peptide of σ -like factor has been shown to function as a chloroplast-targeting signal (20,50). Therefore, there is no doubt that PEP also transcribes photosynthetic genes in the *P.patens* chloroplasts. Our data indicate that the biosynthesis and assembly of the core subunits of PEP are controlled cooperatively by the chloroplast and nuclear genomes, and a relatively more complicated transcription system operates in mosses than in higher plants.

SUPPLEMENTARY MATERIAL

The amino acid sequence alignment of PpRpoA with the α subunit from *E.coli*, *Synechococcus* PCC6301, tobacco and liverwort *M.polymorpha* chloroplasts is available at NAR Online.

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